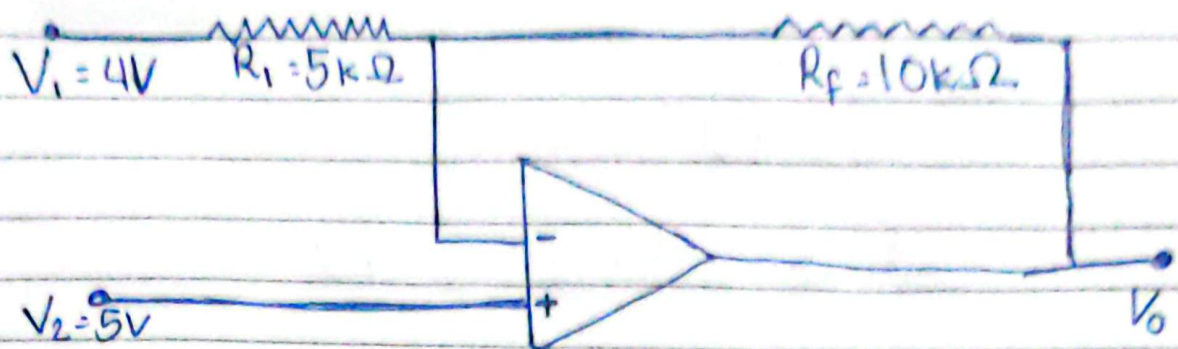


In the subtractor circuit find the value of output voltage (V_o)



Solⁿ

Given:

- (i) Voltage at Non-inverting terminal $V_2 = 5V$
- (ii) $R_1 = 5k\Omega$
- (iii) Voltage at inverting terminal $V_1 = 4V$
- (iv) Feedback resistance (R_f) = $10k\Omega$

In an ideal op Amp

$$V_2 = V_- = 5V$$

~~Applying~~

Current through R_1

$$I = \frac{V_1 - V_-}{R_1}, \quad I = \frac{4 - 5}{5k} = \frac{4 - 5}{10 \times 5 \times 10^3} = -0.2mA$$

Same current flows through R_f

$$\therefore \frac{5 - V_o}{10K} = -0.2mA$$

$$\text{or } 5 - V_o = -2$$

$$\text{or } V_o = 5 + 2$$

$$\therefore V_o = 7V$$

$\therefore$ Output voltage is 7V